

# Implementation of Linear Regression (LR) in Python

## Aim

To implement the **Linear Regression (LR)** algorithm in Python and evaluate its performance by predicting values and calculating the **Mean Squared Error (MSE)** and **R<sup>2</sup> Score**.

## Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
  - NumPy
  - Pandas
  - Scikit-learn (sklearn)

## Theory

Linear Regression is a **supervised machine learning algorithm** used to predict a **continuous dependent variable** based on one or more independent variables. It finds the best-fit straight line that minimizes the prediction error.

### Advantages:

- Simple and easy to implement.
- Fast training and prediction.
- Easy to interpret.

### Disadvantages:

- Assumes a linear relationship between variables.
- Sensitive to outliers.
- Not suitable for complex non-linear data.

## Algorithm

1. Import the required libraries.
  2. Load the Diabetes dataset.
  3. Split the dataset into training and testing sets.
  4. Train the Linear Regression model.
  5. Predict the output for the test data.
  6. Calculate the Mean Squared Error (MSE).
  7. Calculate the  $R^2$  Score.
  8. Display the predicted values and evaluation metrics.
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## Code (Google Colab)

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
data = load_diabetes()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Train Linear Regression model
```

```
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Results
print("First 10 Predicted Values:")
print(y_pred[:10])

print("\nMean Squared Error (MSE):")
print(mean_squared_error(y_test, y_pred))

print("\nR2 Score:")
print(r2_score(y_test, y_pred))
```

## Sample Output

```
*** First 10 Predicted Values:
[138.46970031 181.10052342 125.34400904 292.75977277 123.8830531
 91.89830434 257.26463123 177.76169318 84.98549706 109.15960992]

Mean Squared Error (MSE):
2821.750981001311

R2 Score:
0.4772897164322617
```

## Result

The **Linear Regression** algorithm was successfully implemented in Python using the Diabetes dataset. The model predicted continuous values, and its performance was evaluated using the **Mean Squared Error (MSE)** and **R<sup>2</sup> Score**.